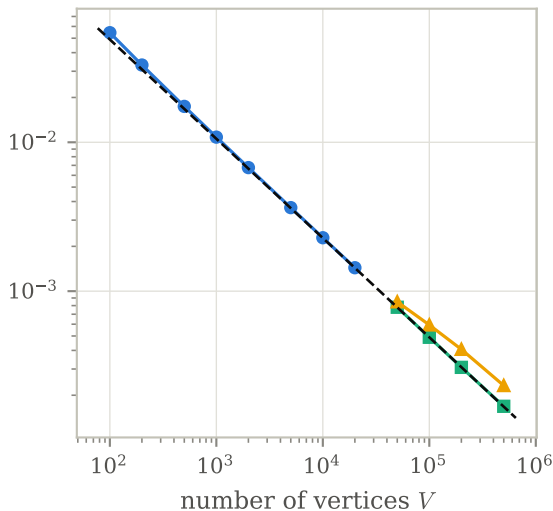
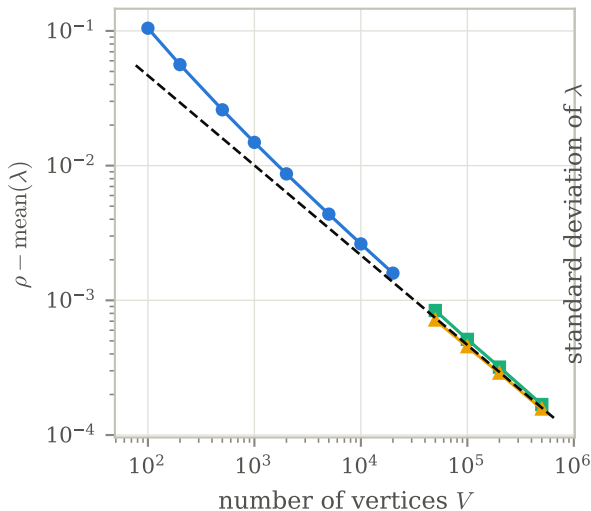


$V^{-2/3}$ scaling of the largest non-trivial eigenvalue — random 4-regular graphs

Gap of the mean below the Ramanujan threshold ρ

Standard deviation



- 4-regular simple (Python) (slope -0.787)
- 4-regular simple (MATLAB) (slope -0.697)
- ▲ 4-regular perm. model (MATLAB) (slope -0.660)
- conjecture $|\mu_{TW_1}| c(4) V^{-2/3}$

- 4-regular simple (Python) (slope -0.684)
- 4-regular simple (MATLAB) (slope -0.666)
- ▲ 4-regular perm. model (MATLAB) (slope -0.562)
- conjecture $\sigma_{TW_1} c(4) V^{-2/3}$